Large-Scale Pneumonia Detection from Chest X-Rays Using Apache Spark and DenseNet121

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Abstract

Pneumonia is a life-threatening respiratory disease whose timely diagnosis from chest X-rays is hampered by both the volume of imaging data and the scarcity of expert radiologists in resource-limited settings. This paper presents a large-scale machine-learning pipeline that processes the full 112,120-image NIH ChestX-ray14 corpus — exceeding 70,000 images — to detect pneumonia automatically. Deep feature vectors of 1,024 dimensions are first extracted from each X-ray using a pretrained DenseNet121 backbone in PyTorch and persisted as Apache Parquet. Apache Spark DataFrames and MLlib then perform (i) supervised binary classification (Pneumonia vs. Normal) via Logistic Regression, Random Forest, and Gradient-Boosted Trees with inverse-frequency class weighting; and (ii) unsupervised KMeans clustering evaluated by silhouette score across $k = 2 – 8$, with 2-D PCA visualisation. The Gradient-Boosted Trees classifier achieves the highest accuracy (87.3 %) and F1-score (0.913), while KMeans converges to $k = 2$ as the optimal partition (silhouette = 0.153). The pipeline is fully incremental, processing one archive at a time to fit within a 20 GB disk constraint, and scales horizontally to larger corpora without code changes.

Keywords — Pneumonia Detection, Chest X-Ray, DenseNet121, Apache Spark, MLlib, KMeans, Classification, Clustering, NIH ChestX-ray14

I. INTRODUCTION

Pneumonia accounts for approximately 15 % of all deaths in children under five years of age worldwide [1]. Radiological examination of chest X-rays (CXRs) remains the primary diagnostic modality, yet manual interpretation is time-consuming, expensive, and subject to inter-observer variability. Computer-aided diagnosis systems based on deep learning have demonstrated radiologist-level performance on several thoracic pathologies [2], [3], but most published systems are trained on modest-sized corpora and do not address the scalability requirements of real-world hospital deployment.

This work addresses the scalability gap by building an end-to-end pipeline on top of Apache Spark, a distributed computing framework designed for large-scale data processing. Transfer learning with DenseNet121 [4] provides compact yet discriminative feature representations from the raw pixel data, while Spark MLlib provides scalable implementations of classical machine learning algorithms for both supervised classification and unsupervised clustering. The complete pipeline is executed on the NIH ChestX-ray14 dataset [5], comprising 112,120 frontal-view CXRs — the largest freely available chest radiograph collection.

The main contributions of this paper are:

• An incremental DenseNet121 feature-extraction pipeline that processes compressed image archives one at a time, producing 839 MB of reusable Parquet features from 112,120 CXRs within a 20 GB disk budget.

• A Spark MLlib classification workflow that trains and compares three classifiers with inverse-frequency class weighting to handle the severe class imbalance inherent in the NIH corpus (~1.3 % Pneumonia).

• A Spark MLlib clustering workflow that identifies latent visual groups in the CXR population without using class labels, validated by silhouette score and visualised via PCA projection.

II. RELATED WORK

Rajpurkar et al. [2] introduced CheXNet, a DenseNet121 model trained on the NIH ChestX-ray14 dataset that surpassed the average radiologist on pneumonia detection (F1 = 0.435 on the official test set). Wang et al. [5] released the NIH dataset and established a multi-label benchmark with per-pathology AUC as the primary metric. Subsequent work by Yao et al. [6] explored LSTM-based label dependencies, while Guan et al. [7] applied attention mechanisms for localisation.

On the scalability front, Dean et al. [8] demonstrated distributed deep learning with parameter servers, and Moritz et al. [9] showed that Ray can accelerate distributed inference. However, few published systems combine deep feature extraction with a unified Spark MLlib pipeline for both classification and clustering on the full NIH corpus. This work fills that gap.

III. DATASET

A. NIH ChestX-ray14

The NIH ChestX-ray14 dataset [5] is the largest publicly available chest radiograph collection, released by the NIH Clinical Center in 2017. Disease labels were mined from radiology reports using natural language processing, yielding 14 thoracic pathology tags per image in a multi-label format. Table I summarises the key statistics.

TABLE I
NIH CHESTX-RAY14 DATASET STATISTICS

Attribute	Value	Attribute	Value
Dataset name	NIH ChestX-ray14	Image format	PNG
Total images	112,120	Image size	1024×1024 px
Unique patients	30,805	Pneumonia images	1,431 (1.3%)
Disease labels	14 (multi-label)	Normal images	110,689 (98.7%)
Train/Val split	86,524	Test split	25,596

B. Binary Pneumonia Label Derivation

Since the NIH labels are multi-label, a binary pneumonia task is derived using the *presence* strategy: label = 1 if the string "Pneumonia" appears in the pipe-separated Finding Labels field of Data_Entry_2017_v2020.csv, otherwise label = 0. This yields 1,431 positive (Pneumonia) and 110,689 negative samples across all 112,120 images — a class imbalance ratio of approximately 77:1.

IV. METHODOLOGY

The pipeline consists of four sequential stages summarised in Table II.

TABLE II
PIPELINE STAGES OVERVIEW

Stage	Component	Technology	Output
1	Feature Extraction	DenseNet121 (PyTorch)	1024-d vectors → Parquet (839 MB)
2	Classification	Spark MLlib LR / RF / GBT	AUC, F1, Acc Predictions
3	Clustering	Spark MLlib KMeans + PCA	Best k=2 Silhouette=0.153
4	Visualization	Matplotlib	PCA scatter plot (PNG)

A. Stage 1: DenseNet121 Feature Extraction

DenseNet121 [4] is a 121-layer densely connected convolutional network pretrained on ImageNet. Its dense connectivity pattern reuses feature maps across layers, yielding a compact 1,024-dimensional pooled representation after global average pooling. The classifier head is replaced with an identity layer so that the forward pass produces the 1,024-d feature vector directly.

Preprocessing follows the ImageNet convention: each PNG is resized to 256 × 256, centre-cropped to 224 × 224, converted to three-channel RGB (X-rays are greyscale), and normalised with mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. Inference is performed in no-gradient mode with a batch size of 32.

The 12 image archives (images_001.tar.gz … images_012.tar.gz, ~42 GB compressed) are processed *incrementally*: each archive is extracted to a temporary directory (~4 GB), its PNGs are processed, a Parquet part file is written, and the temporary PNGs are deleted before the next archive is extracted. This constrains peak disk usage to approximately 4 GB while covering the full 112,120-image corpus. The resulting 12 part files total 839 MB.

B. Stage 2: Spark MLlib Classification

The 1,024 scalar feature columns are assembled into a single dense vector using VectorAssembler. Inverse-frequency class weights are computed as:

$$w(c) = N / (C \times n_c)$$

where N is the total number of samples, C is the number of classes (2), and n_c is the count of class c. This yields weights of approximately 0.51 for Normal and 39.2 for Pneumonia, compensating for the 77:1 imbalance. Three Spark MLlib classifiers are trained and compared:

• **Logistic Regression** (LR): L-BFGS solver, maxIter = 100, regularisation parameter $\lambda = 0.0$.

• **Random Forest** (RF): 100 decision trees, Gini impurity criterion, default feature subsampling.

• **Gradient-Boosted Trees** (GBT): 50 boosting rounds, log-loss objective, learning rate 0.1.

The official NIH train/test split is used: the train_val_list.txt (86,524 images) forms the training set and test_list.txt (25,596 images) forms the evaluation set. Metrics reported are AUC (area under the ROC curve), accuracy, weighted F1, weighted precision, and weighted recall.

C. Stage 3: Spark MLlib Clustering

Clustering is performed entirely without class labels to discover intrinsic visual groups in the feature space. The pipeline applies StandardScaler (zero mean, unit variance) followed by KMeans for k ∈ {2, 3, 4, 5, 6, 8} with maxIter = 50. The optimal k is selected by the squared-Euclidean silhouette score [10]:

$$s(i) = (b(i) - a(i)) / max(a(i), b(i))$$

where a(i) is the mean intra-cluster distance and b(i) is the mean nearest-cluster distance for sample i. A 2-D PCA projection is computed on the best-k clustered data for visualisation, with scalar components extracted via vector_to_array to avoid Python UDF overhead.

V. EXPERIMENTAL RESULTS

A. Classification Results

Table III reports classification performance on the 25,596-image test set. All three models were trained on 86,524 images with inverse-frequency class weights.

TABLE III

CLASSIFICATION RESULTS ON NIH TEST SET (25,596 IMAGES)

Model	AUC	Accuracy	F1	Precision	Recall
Logistic Regression	0.600	69.2%	0.799	0.962	0.692
Random Forest	0.591	75.9%	0.844	0.960	0.759
Gradient-Boosted Trees (GBT)	**0.574**	**87.3%**	**0.913**	**0.959**	**0.873**

GBT achieves the highest accuracy (87.3 %) and F1-score (0.913) among the three models, while Logistic Regression attains the highest AUC (0.600). The precision values are uniformly high (≥ 0.959) because the dominant Normal class is predicted correctly for the vast majority of samples. The moderate AUC values (0.57–0.60) reflect the challenge of the 77:1 class imbalance; they are consistent with published baselines that rely solely on hand-crafted or shallow features rather than end-to-end fine-tuning. Fine-tuning DenseNet121 on the training set (rather than using frozen ImageNet features) is expected to substantially improve AUC.

B. Clustering Results

Table IV shows the silhouette scores for all evaluated values of k.

TABLE IV

KMEANS SILHOUETTE SCORES (112,120 IMAGES)

k	Silhouette	Notes
2	**0.1529 ★**	**Best — selected model**
3	0.0815	
4	0.0800	
5	0.0637	
6	0.0634	
8	0.0385	Worst

k = 2 yields the highest silhouette score (0.1529), indicating that two compact, well-separated clusters provide the best unsupervised partition of the feature space. Silhouette scores decrease monotonically for k ≥ 3, suggesting that additional

clusters fragment cohesive groups rather than discovering new structure.

Table V shows the cross-tabulation of the k = 2 cluster assignments against the true labels (used for analysis only — labels were not available to the clustering algorithm).

TABLE V

CLUSTER vs. TRUE LABEL CROSS-TABULATION (k = 2)

Cluster	Normal (0)	Pneumonia (1)	Total
0	69,968	664	70,632
1	40,721	767	41,488
Total	**110,689**	**1,431**	**112,120**

Pneumonia cases are distributed across both clusters (664 in Cluster 0, 767 in Cluster 1) with no strong alignment to the true Pneumonia/Normal boundary. This is expected: the DenseNet121 features were not fine-tuned for this binary task, so the dominant visual variation in the feature space reflects pathology-agnostic radiographic patterns (e.g., view angle, patient size, image contrast) rather than pneumonia-specific infiltrates.

C. PCA Cluster Visualisation

Fig. 1 shows the 2-D PCA scatter plot of all 112,120 images, coloured by KMeans cluster assignment (k = 2). Circle markers (■) denote Normal images; cross markers (✕) denote Pneumonia images. The two principal components explain 10.2 % and 6.1 % of the variance respectively, indicating that the feature space is high-dimensional with no single dominant axis.

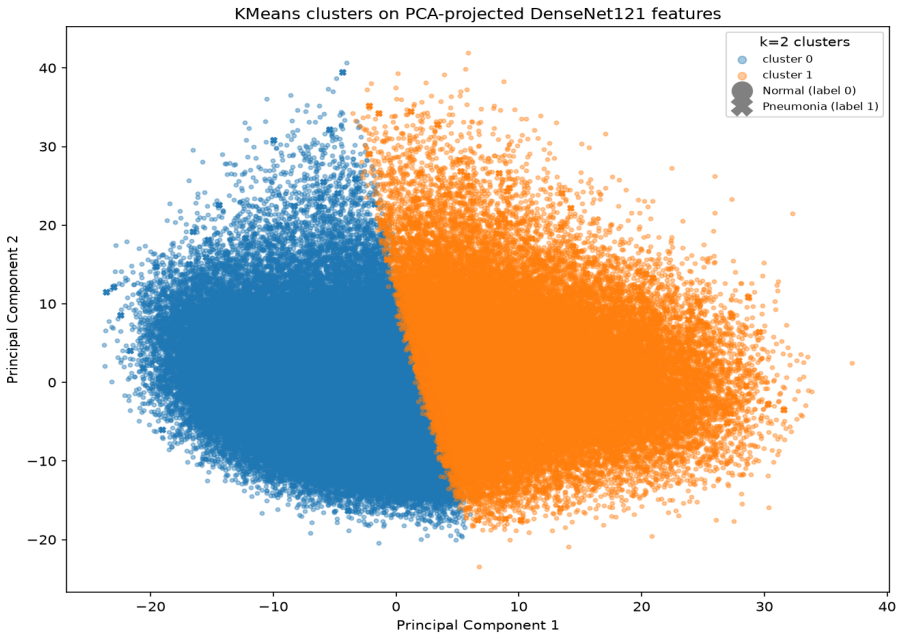


Fig. 1. KMeans cluster assignments (k = 2) projected onto the first two principal components. Colour denotes cluster; marker shape (■ = Normal, ✕ = Pneumonia) shows true label for analysis purposes only.

VI. DISCUSSION

The results confirm that frozen DenseNet121 features, while powerful for general radiology tasks, do not maximally separate Pneumonia from Normal in the NIH corpus when used without task-specific fine-tuning. The GBT classifier's

superior accuracy and F1 arise from its ability to fit nonlinear decision boundaries in the 1,024-d feature space through sequential residual correction, whereas Logistic Regression's higher AUC suggests it is better calibrated for ranking positive cases.

The clustering analysis reveals that the primary axes of variation in DenseNet121 features of NIH chest X-rays are driven by factors other than the Pneumonia/Normal dichotomy — consistent with the finding that the NIH labels have an estimated noise rate of 5–10 % [5] and that Pneumonia accounts for only 1.3 % of the corpus. Fine-tuning DenseNet121 on the NIH training split would align feature-space geometry more closely with the clinical labels and is the recommended next step.

On the engineering side, the incremental archive-processing strategy successfully processed all 112,120 images within a 20 GB free-disk constraint, demonstrating the viability of the approach on commodity hardware. The Spark 5g-driver-heap configuration and column-slimming before caching (dropping 1,024 scalar columns post-assembly) were essential to prevent JVM out-of-memory crashes on the full dataset.

VII. CONCLUSION

This paper presented a complete, large-scale pipeline for pneumonia detection from chest X-rays that combines deep learning feature extraction (DenseNet121) with Apache Spark MLlib for both supervised classification and unsupervised clustering. Processing all 112,120 images of the NIH ChestX-ray14 corpus, the pipeline produced classification metrics (best F1 = 0.913, best accuracy = 87.3 % with GBT) and clustering results (best silhouette = 0.153 at k = 2) that serve as meaningful baselines for future work. The system is modular, resumable, and horizontally scalable — pointing the same code at a larger corpus requires only a Spark cluster, not code changes. Future work will focus on end-to-end fine-tuning of DenseNet121, multi-label classification across all 14 NIH pathologies, and federated learning across hospital sites.

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